

Original Research

Extracting biomedical relation from cross-sentence text using syntactic dependency graph attention network

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ARTICLE INFO

Keywords:

Biomedical relation extraction
Cross-sentence text
Feature engineering
Graph-oriented
Syntactic dependency graph attention

ABSTRACT

In biomedical literature, cross-sentence texts can usually express rich knowledge, and extracting the interaction relation between entities from cross-sentence texts is of great significance to biomedical research. However, compared with single sentence, cross-sentence text has a longer sequence length, so the research on cross-sentence text information extraction should focus more on learning the context dependency structural information. Nowadays, it is still a challenge to handle global dependencies and structural information of long sequences effectively, and graph-oriented modeling methods have received more and more attention recently. In this paper, we propose a new graph attention network guided by syntactic dependency relationship (SR-GAT) for extracting biomedical relation from the cross-sentence text. It allows each node to pay attention to other nodes in its neighborhood, regardless of the sequence length. The attention weight between nodes is given by a syntactic relation graph probability network (SR-GPR), which encodes the syntactic dependency between nodes and guides the graph attention mechanism to learn information about the dependency structure. The learned feature representation retains information about the node-to-node syntactic dependency, and can further discover global dependencies effectively. The experimental results demonstrate on a publicly available biomedical dataset that, our method achieves state-of-the-art performance while requiring significantly less computational resources. Specifically, in the “drug-mutation” relation extraction task, our method achieves an advanced accuracy of 93.78% for binary classification and 92.14% for multi-classification. In the “drug-gene-mutation” relation extraction task, our method achieves an advanced accuracy of 93.22% for binary classification and 92.28% for multi-classification. Across all relation extraction tasks, our method improves accuracy by an average of 0.49% compared to the existing best model. Furthermore, our method achieved an accuracy of 69.5% in text classification, surpassing most existing models, demonstrating its robustness in generalization across different domains without additional fine-tuning.

1. Introduction

As the volume of biomedical literature grows rapidly, it becomes increasingly difficult and costly to manually extract the biomedical knowledge. Fortunately, text mining technology can uncover this potential knowledge quickly and accurately. Relation extraction is one of the important tasks in text mining, which can help medical researchers to retrieve the association information in the literature efficiently.

Relation extraction methods for biomedical information mining are of great significance, which can help researchers to quickly integrate potential knowledge from a large amount of literature [1]. Therefore, research on low-cost, efficient biomedical relation extraction methods has received increasing attention.

The relation extraction task can be defined as a problem for classifying the relation classes of given sentences and entities. The common relation extraction task is the single-sentence binary relation extraction,

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<https://doi.org/10.1016/j.jbi.2023.104445>

Received 7 January 2023; Received in revised form 6 June 2023; Accepted 15 July 2023

Available online 17 July 2023

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which is limited to extracting relations between two entities and requires them to be in the same sentence. However, in biomedical literature, the relations are usually expressed by multiple sentences together and involve more entities. Considering the example in Table 1, these two sentences involve three types of entities: gene, drug, and mutation, and jointly express the relation between them, i.e., “The L858E mutation in the EGFR gene responds to treatment with gefitinib drug.”. Obviously, this relation involves multiple entities, and each of these sentences cannot separately express the relation correctly. For the biomedical literature, longer texts often can express richer information (e.g., specific effects of the drug). Therefore, the cross-sentence relation extraction is proposed and receives more and more attention. Extracting biomedical relation from cross-sentence texts can help medical researchers to obtain more valuable knowledge, which is of great significance for biomedical research.

It is very important to learn the feature representation of sentences effectively to improve the performance of relation extraction. In particular, in cross-sentence texts that involve multiple sentences, where the sequence length is often much longer than that of a single sentence, it is more difficult to learn effective feature representation. Common recursive models such as RNN and LSTM have fixed sequential computation constraints, which hinder parallelization and make them inefficient in handling long sequence problems. Fortunately, the attention mechanism allows parallel computation and can access the entire sequence at each time step, effectively reducing the computational cost of modeling long sequences. However, when the sequence is too long, the computational complexity of the attention mechanism is high, and it is difficult to learn accurate global dependencies and structural information. Further, we show an example in Appendix A that describes the operation of the recursive method, the self-attention method, and the SR-GAT method (ours).

Recently, when dealing with long sequence problems, most studies usually parse the input from sequence into document graph, and then learn its feature representation using graph structure-oriented feature engineering methods [2]. These methods introduce syntactic structure as prior knowledge into the modeling process, which has significant advantages for capturing global dependency information of long sequences. However, most of the existing methods for graph structure-oriented modeling are complex in structure and high in operational costs. Currently, it is still a challenging problem to learn feature representation of long sequences by processing large amount of cross-sentence information efficiently and at low-cost.

To address this challenge, in this paper, from the aspect of feature engineering, we propose a novel graph attention network guided by syntactic dependency relationships (SR-GAT) for learning feature representation of long sequences, which can be used to extract biomedical relation from cross-sentence texts. Our method is oriented towards graph structure modeling, which allows each node to focus on all nodes in its neighborhood without the limitation of sequence length, so it can effectively solve the problem of overly long sequence in cross-sentence. Furthermore, syntactic dependency graph contains dependency information between nodes, so we propose a graph probability network that encodes syntactic dependency information (SR-GPR) to guide graph attention network to focusing on structural information about sequences. In the SR-GAT network, the feature of each node is aggregated from the information of all nodes in its neighborhood, and the syntactic

dependency encoded by the SR-GPR network are used to control the degree of aggregation between nodes, so that it can effectively learn the global dependence and structural information of long sequences.

To summarize, our contributions are as follows:

- We propose a general feature engineering method, where graph structure modeling enables it to solve the problem of overly long sequences, and syntactic dependency encoding enables it to learn global dependencies and structural information efficiently. Moreover, we applied it to extract biomedical relation from cross-sentence texts, which can help researchers to obtain richer knowledge quickly and efficiently.
- The proposed method can be able to quickly learn structural information of long sequences just by a simple network structure, where encoding syntactic dependency is used to guide the aggregation of information between nodes in the same neighborhood.
- Experimental results show that our method achieves state-of-the-art performance in biomedical relation extraction task and can be robustly generalized to other tasks without additional fine-tunings.
- Our code is publicly available for future research, which is available at: <https://github.com/Zxy-MLlab/SR-GAT>.

2. Related works

Deep learning-based methods for single-sentence binary relation extraction have been widely studied with great success. In the biomedical field, such relations include gene-protein expression, drug-drug interactions [3], drug-disease interactions and so on. Common methods are sequence models based on RNN, LSTM [4], Self-Attention [5] and convolutional network [67] are used for the binary relation extraction task. Recently, many researchers have used pre-trained language models to accomplish fine-tuning of binary relation extraction tasks [89]. Such methods do not require redesigning the network structure or excessive training data, and can exhibit competitive performance to some extent. However, these methods are only applicable to short sequences with simple sentence structures, and as the sequence length increases, the required network structure becomes more and more complex, with the increasing training cost, and it is difficult to learn the feature representation that contains the global dependency information.

In recent years, the research on cross-sentence text has received more and more attention [1011]. Cross-sentence relation extraction helps researchers to extract richer knowledge from the literature data, and therefore has received more and more attention [12]. Peng et al. [13] firstly proposed a graph structure-oriented LSTM model. They used the Stanford syntactic parser to parse the input sequence into two directed dependency graphs, and used two LSTM models to extract features from the left and right directed graphs respectively, which can capture the dependency information of long sequences for cross-sentence relation extraction. Song et al. [14] improved on Peng’s method and proposed the GS GLSTM model. They designed a message passing mechanism that was able to maintain the original graph structure for solving the problem of information loss. Secondly, by passing information to each other, the information of neighboring nodes can be well integrated, while achieving more parallel computation and improving the efficiency of the model.

Recently, Guo et al. [15] proposed an attention-guided graph convolutional network model (AGGCN) in which they use intensive connectivity to integrate information across sentences, allowing each token to receive information from all tokens in the previous sublayer. Zhao et al. [16] proposed an N-gram graph LSTM network oriented to graph structure modeling, which makes full use of the information of graph structure and solves the problem of high depends on graph structure and lack of context sequence structure information in GNN. Zhao et al. [17] proposed a relation extraction method based on multi-headed attention and graphical convolutional networks, where graphical convolutional

Table 1
An example of biomedical relation in cross-sentence text.

Sentences	a)	The deletion mutation on exon-19 of <i>EGFR</i> gene was present in 16 patients, while the <i>L858E</i> point mutation on exon-21 was noted in 10.
	b)	All patients were treated with <i>gefitinib</i> and showed a partial response.
Relation	Response	

networks were used to encode dependency information for guiding multi-headed attention to learn actual dependency information, improving the performance of cross-sentence relation extraction in biomedicine. Lai et al. [18] added the graph Transformer structure to the model of BERT and proposed the BERT-GT framework, which achieved the most competitive performance in the biomedical relation extraction task.

Previous studies have shown that attention mechanism is effective in learning the feature representation. Velickovic et al. [19] generalized the attention mechanism to the graph structure and proposed a graph attention network (GAT) for learning the feature representation of the graph structure. Wang et al. [20] introduced additional relational heads to extend the original GAT and proposed a relational graph attention network (R-GAT) for the sentiment analysis task. They use two types of multi-headed attention mechanisms (attention head and relation head) to learn the feature representation of the input graph, achieving competitive performance in the sentiment analysis task.

In this paper, we propose a novel graph attention network guided by syntactic dependency graph for extracting biomedical relation from cross-sentence text. Compared with GAT, our method encodes syntactic dependency information in the input graph, aggregates dependency information in the neighborhood nodes, and is able to better learn the structural information of long sequences. Compared to R-GAT, our method focuses on learning the dependency information, and the attention between nodes is only derived by encoding the syntactic edges between them without considering the node information. From this perspective, our method is simpler and has lower computational overhead, and more importantly, experimental results show that our method has state-of-art performance in the task of extracting drug-utility relations from cross-sentence texts.

3. Problem definition

Essentially, cross-sentence relation extraction can be represented as a classification problem. Specifically, given a document consisting of multiple sentences, the labeled entities in the document, and a pre-defined logical relation set. Our goal is to analyze the specific logical relations between entities expressed in given document. In particular, entities may be distributed in different sentences.

Traditional relation extraction tasks often require entities to be in the same sentence and mostly are binary relation, such as single-sentence binary relation extraction. For complex relations, however, it is often rare that related entities appear in a sentence at the same time, while the number of entities involved in complex relations increases from two to more. In particular, in biomedical research, cross-sentence relation extraction can help researchers obtain richer knowledge. As can be seen from Table 1, the single sentence binary relation extraction can only detect that a certain type of patient responds to the treatment with “gefitinib” from sentence b), but it cannot determine the type of disease the drug can treat, because the patient’s information is mentioned in previous sentence a). Differently, the cross-sentence relation extraction can directly detect the exact drug effect: i.e., “The “L858E” mutation in the “EGFR” gene responds to treatment with the “gefitinib” drug”.

4. Methods

In this paper, we use the syntactic dependency to guide the graph attention mechanism for learning structural information of long sequences, and propose a novel graph attention network (SR-GAT). In traditional GAT networks, the features of each node are aggregated by nodes with which it has edges, but the whole process does not consider syntactic dependency, which may lose important structural information. To address this problem, we propose a graph probability network for encoding syntactic dependency graph (SR-GPR), which maps syntactic information between nodes to the degree of aggregation between them. In the SR-GAT network, the feature of each node is generated by

aggregating the information of all nodes in its neighborhood, and the aggregation degree is given by the SR-GPR network, which encodes the global dependency and structural information of the input graph. The feature of the syntactic dependency graph will be concatenated with the entities feature to form the input of the classifier, which will be used for relation classification. We show our model’s overall architecture in Appendix B.

4.1. Syntactic dependency graph

In the cross-sentence relation extraction task, the relevant entities are often not in the same sentence, so it is very important to learn dependencies between contexts and the sentence structure accurately. A typical method is to replace the input from a sequence representation of the sentence into a syntactic dependency graph. The syntactic dependency graph consists of word nodes, dependent edges, neighboring edges of adjacent words and associations between sentences, which contains syntactic dependency and structural information for achieving a better characterization of structural features between cross-sentence.

In this paper, we refer to Peng et al. [13] and represent the cross-sentence text as a syntactic dependency graph that integrates intra-sentence and inter-sentence dependencies. Further, we provide an example of the syntactic dependency graph in Appendix C.

4.2. Syntactic dependency graph probability network

Syntactic dependency contains the structure information, which is crucial when dealing with long sequence problems. Therefore, it should be used to guide the graph attention mechanism to aggregate information about nodes in the neighborhood. In this paper, for accomplishing this task, we propose a syntactic dependency graph probability network (SR-GPR) to encode syntactic edges in graph, as shown in Fig. 1.

In Fig. 1, the input of the SR-GPR network is a syntactic dependency graph, which is preprocessed to obtain an edge set and a mask matrix. The former is used to represent the syntactic dependency between nodes in the input graph, which can be viewed as a connected graph and will be passed on to the convolutional layer. The latter is passed directly to the mask layer. The output of SR-GPR network is a correlation probability matrix, which represents the correlation probability between nodes. We will introduce it in detail as follows.

For any edge in the edge set, it connects two nodes and indicates the syntactic dependency between them. We first transform specific syntactic dependency into the vector representation by word embedding. Therefore, the syntactic edge set can be viewed as a multi-channel

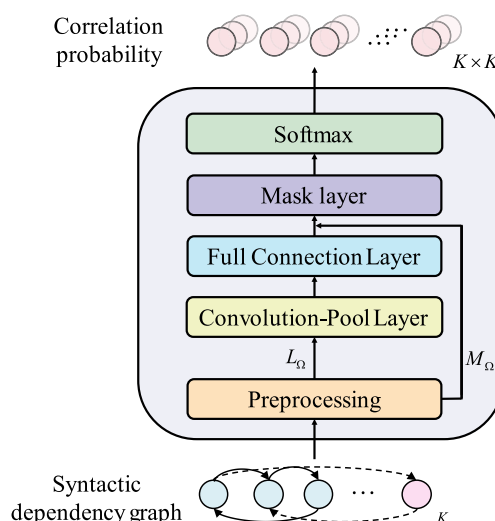


Fig. 1. Syntactic Dependency Graph Probability Network (SR-GPR Network).

image, on we perform 2D convolution to capture the information of dependency between nodes. Further, we aggregate the features using 2D max-pooling to reduce the dimension. Finally, we use a fully connected layer to map the dependency features between nodes to the correlation coefficient between them.

It is worth noting that, the number of syntactic edges connected to each node is different. However, in the previous operation, we expand it to the maximum number of nodes, which will inevitably introduce wrong signals and create unnecessary interference to the information aggregation of nodes. To solve this problem, we introduce a Mask layer, which is used to mask these meaningless virtual edges. As can be seen in Fig. 1, the mask matrix is given directly by the preprocessing module, which has the same dimension size as the syntactic edge set, and is used to indicate whether the syntactic dependency between the nodes really exists. If there is a syntactic edge between two nodes, we set the element at the corresponding position in the mask matrix to 1, otherwise to 0. We then multiply it with the correlation probability matrix between nodes by element-wise multiplication to mask the correlation coefficient calculated from the virtual edges.

Finally, we use a softmax function to normalize the correlation coefficient matrix, ensuring that the nodes associated with each node are assigned a sum of weights to 1, and to obtain the correlation probability matrix.

4.3. Syntactic dependency graph attention network

The input to the cross-sentence relation extraction task contains multiple sentences, which requires that the model must have the ability to handle long sequences. Compared with the sequence structure, syntactic dependency graphs contain richer structural information, so graph structure-oriented modeling methods usually have an advantage over sequence-oriented modeling methods in dealing with long sequence problems. In this paper, we propose a syntactic dependency graph attention network (SR-GAT) that aggregates information about all nodes in its neighborhood for each node under the guidance of the SR-GPR network, as shown in Fig. 2.

In the SR-GAT network, the feature of each node is derived by weighting the original nodes associated with it, where the weights are

given by SR-GPR. Specifically, given a syntactic dependency graph, on the one hand, we directly feed it into the SR-GPR network to calculate the correlation probability matrix between the nodes. On the other hand, we do word embedding on the nodes in the graph to obtain the vector representation of each word. We then multiply it with the correlation probability matrix by matrix multiplication to aggregate the information of the relevant nodes.

Furthermore, in order to learn more complex structural information and capture deeper logical relations, we use multiple sets of SR-GAT based on independent parameter spaces to calculate the feature representations. Finally, we use the average method to summarize them to obtain the final output feature. In SR-GAT, the information aggregation between nodes is guided by the syntactic dependency graph, which enables it to learn the structural information of long sequences quickly and accurately.

4.4. Task network

The input of the task network consists of the output of the SR-GAT network and the encoding feature of entities involved in the relation to be extracted. The former encodes the structural information of the sentences, which is used to better capture the long-term dependencies between distant entities. The latter directly encodes the features of entities, which is indispensable in relation extraction tasks. In particular, in the cross-sentence relation extraction task, a relation example usually involves multiple entities. In this paper, we fix the dimension of entity features and aggregate the features of all entities by using the average method. This makes our method to be easily generalized to any number of entities for relation extraction.

In the task network, we use a convolutional layer and a maximum pooling layer to complete the dimensionality reduction of the input features, and then maps the features to the pre-defined relation set through two fully connected layers and a softmax layer. (The task network structure graph is shown in Appendix D).

5. Experiments and results

5.1. Data preparation

For the cross-sentence relation extraction task, we evaluate our model by using the dataset from Peng et al. [13]. This is a biomedical dataset from the PubMed database, covering relations between “drug”, “gene”, and “mutation” entities, including 6987 instances of ternary relations (“drug-gene-mutation”) and 6087 instances of binary relations (“mutation-drug”). There are five class of relations in this dataset: “resistance or non-responsive”, “sensitivity”, “response”, “resistance” and “None”.

In this paper, we will test the extraction performance of our model on this dataset for five classes of relations between “drug-gene-mutation” entities and five classes of relations between “mutation-drug” entities. Furthermore, we follow Peng et al. [13] to treat all related labels as “Yes” and treat “None” as “No” to make the multiple classification labels binarized. We test the extraction performance of our model for binary relation between “drug-gene-mutation” entities and “mutation-drug” entities.

In addition, to further validate the effectiveness of the SR-GAT network in extracting features, we use the Corpus of Linguistic Acceptability (CoLA) dataset to evaluate its performance in the text classification task. This dataset consists of 10,657 sentences from 23 linguistic publications, including 6,338 positive samples and 2,690 negative samples. The given task is to determine whether the given sentence is grammatically correct or not, so it can be considered as a single-sentence text classification problem.

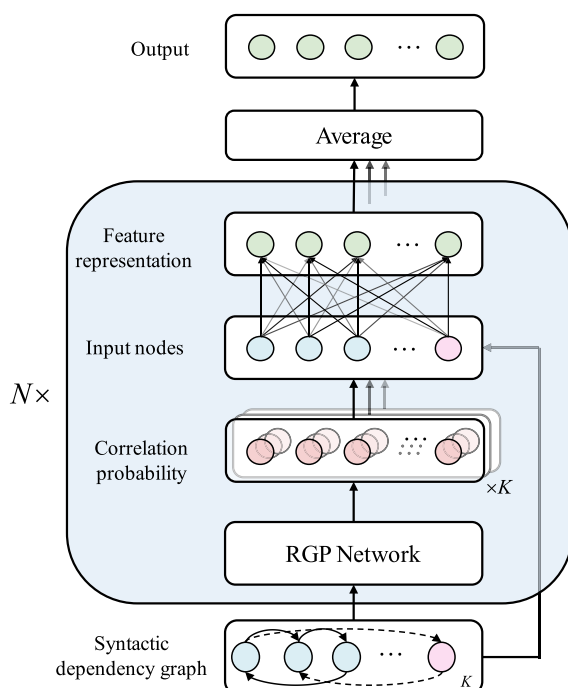


Fig. 2. Syntactic Dependency Graph Attention Network (SR-GAT Network).

5.2. Parameter settings

In experiments, we follow the strategy of Peng et al. [13] and evaluate our model using five-fold cross-validation, while calculating the final performance of the model by averaging the accuracies over five independent test sets.

During training, word embeddings are initialized using the publicly available 100-dimensional GloVe vectors [21]. The edge embedding is initialized using random 50-dimensional vectors. The pre-loaded word embedding vectors are not updated during the model training. The maximum sentence length is set to 128 and the hidden state dimension of the fully connected layer is set to 512.

In the SR-GPR networks, we use 3 independent sets of parameter spaces to learn the feature representation of sentence structure respectively. To alleviate the overfitting problem, we used the dropout technique in different layers and set the dropout rate to 0.6. The Adam optimization algorithm is used and the learning rate is set to 0.0001. All parameter settings are shown in Appendix E.

5.3. Experimental results

5.3.1. Results on biomedical cross-sentence relation extraction

We present the performance of our method compared with other typical methods for the cross-sentence relation extraction task in Table 2. As can be seen in Table 2, the deep learning-based methods mostly outperform the direct feature-based methods. The Bi-LSTM-based method outperforms the CNN-based method, which also shows the greater advantage of LSTM in processing long sequences. The

Table 2

Average test accuracy of the five-fold cross-validation on the dataset. Two-class means “Yes” and “No” binary relation, and Multi-class means “resistance or non-responsive”, “sensitivity”, “response”, “resistance” and “None” five-class relation. The bold in the table means the best score.

Methods	Drug-Mutation		Drug-Gene-Mutation	
	Two-class	Multi-class	Two-class	Multi-class
BLSTM (Zhang et al. [22], 2015)	79.00	–	78.90	–
Multi-Att-CNN (Wang et al. [23], 2016)	78.90	–	79.70	–
Att-BLSTM (Zhou et al. [24], 2016)	79.80	–	79.90	–
FEATURE-BASED (Quirk et al. [25], 2017)	75.20	–	77.70	–
CNN (Peng et al. [13], 2017)	74.90	–	78.10	–
Bi-LSTM (Peng et al. [13], 2017)	76.00	–	80.10	–
Bi-LSTM MULTI-TASK (Peng et al. [13], 2017)	78.10	–	82.40	–
GRAPH LSTM-EMBED (Peng et al. [13], 2017)	76.50	–	80.60	–
GRAPH LSTM-FULL (Peng et al. [13], 2017)	76.70	–	80.70	–
GRAPH LSTM MULTI-TASK (Peng et al. [13], 2017)	78.50	–	82.00	–
Bidir DAG LSTM (Song et al. [14], 2018)	76.40	50.70	77.30	51.70
GS GLSTM (Song et al. [14], 2018)	88.56	86.92	87.60	85.91
GCN (Full Tree) (Zhang et al. [26], 2018)	83.60	74.30	84.80	77.50
LSTM-CNN (Mandya et al. [27], 2019)	88.50	–	82.90	–
AGGCN (Guo et al. [15], 2019)	85.36	76.88	86.44	79.62
BERT (Devlin et al. [5], 2019)	91.01	90.03	90.80	90.73
BlueBERT (Peng et al. [28], 2019)	92.10	92.16	91.27	91.10
LF-GCN (Guo et al. [29], 2020)	87.10	–	88.40	–
GA LSTM (Zhao et al. [16], 2020)	85.80	77.50	87.30	83.80
NGG LSTM (Zhao et al. [16], 2020)	86.50	77.80	88.40	84.40
MA-GCN (Zhao et al. [17], 2021)	91.30	–	87.80	–
BERT-GT (Lai et al. [18], 2021)	93.27	93.45	91.51	91.35
SR-GAT (Ours)	93.78	92.14	93.22	92.28

methods based on CNN or LSTM with added attention mechanism are better than using them alone. The hybrid method based on LSTM-CNN outperforms the method using either of them alone due to its ability to learn locally important features while capturing long-term dependencies of sequences. On the other hand, the graph structure-based modeling methods outperform the sequence-based modeling methods. The GRAPH-LSTM extends LSTM to graph structures, combining various intra- and inter-sentence dependencies, and is able to encode rich structure information, which improves the performance of the model. The GS GLSTM improves the GRAPH-LSTM to realize higher parallelism and also has better performance. The graph attention-based LSTM (GA LSTM) and N-gram graph LSTM (NGG LSTM) methods show competitive performance in handling two-classification task. It is worth noting that, the AGGCN requires the dense connection for information aggregation, and performs well in extracting binary relation, but performs poorly in extracting five-classification relation. It is also clear that, both the BERT-based and BERT-improved methods are able to exhibit more than 90% performance, thanks to its Self-Attention mechanism. The BERT-GT is a hybrid model based on BERT and Graph Transformer, which enables the combination of BERT and graph structure to achieve the best performance in binary multi-classification relation extraction task.

In summary, our method is oriented to syntactic dependency graph modeling, and achieves advanced performance on different relation extraction tasks. Our method is a significant improvement over GRAPH LSTM and GS GLSTM, and the attention mechanism allows for more parallelization, which makes our network more efficient. Compared with the method based on BERT, our method is also significantly competitive, especially our method has a simple network structure and low computational overhead, which is of great significance when dealing with complex information extraction tasks.

In addition, we report performance ranges (min, median, max) of our method in five-fold cross-validation, and the results are shown in Appendix F, which further proves the effectiveness of our method.

5.3.2. Computational complexity and time cost analysis

As the information extraction task become more and more complex, it is critical to reduce computational overhead and training costs. Compared with the recurrent model, the attention mechanism breaks through the basic constraint of sequential computation and significantly improves the computational efficiency of the model by allowing more parallel computation. In this paper, our method is based on the self-attention mechanism, and the computational complexity of the model is further reduced by the guidance of the syntactic dependency graph. The study by Vaswani et al. shows that the computational complexity of information aggregation of self-attention mechanism is $O(\bar{k}^2 \cdot d)$, where $\bar{k}$ is the length of the input sequence and d is the dimension of the feature. For our method, the information aggregation of each node only focuses on the nodes in its neighborhood, not all nodes. Therefore, the computational complexity of information aggregation is $O(\bar{k} \cdot \bar{q} \cdot d)$, where $\bar{q}$ is the average number of nodes' edges in the syntactic dependency graph. According to our statistics, in the data set proposed by Peng et al., the average sentence length $\bar{k}$ is 66.47, while the average number of edges $\bar{q}$ is 3.89. Obviously, our method has a lower computational complexity and has a greater advantage as the sequence length increases.

In addition, the trainable parameter size and training time are also

Table 3

Model computational complexity and time cost.

Evaluation metrics	Value	Unit
Parameters	281	k
Training Cost (FLOPs)	3.9	10^9
Training time	223	s
Epoch number	100	–

important metrics to measure model performance in practical applications. In Table 3, we report the number of the parameter, training cost, training time and training epoch number of our model. It is reported in Table 2 that most Bert-based methods achieve high accuracy, but even the basic BERT model has 110000 k parameters, much larger than our model (281 k). Furthermore, our model has an acceptable training time in one epoch, and can converge within 100 rounds, which also shows advantages and potentials in practical applications.

5.3.3. What does the SR-GAT network pay attention to?

The SR-GAT network proposed in this paper is oriented to syntactic graph modeling, which can effectively learn the feature representation of long sequences. Compared with the traditional self-attention mechanism, it focuses on a small number of important syntactic dependency. To explore what information the SR-GAT network actually pays attention to, we show an example in Fig. 3. The syntactic dependency graph for this example is shown in Fig. 3 (a). For simplicity, we only show some syntactic edges that are important for relation extraction. Fig. 3 (b) shows the attention weights between the nodes in this example.

This example presents the fact that “v600e allele is resistant to the efficacy of the panitumumab drug”. From Fig. 3 (b), it can be seen that “impaired” and “effect”, as well as “effect” and “cetuximab”, are not adjacent but have semantic connections. Therefore, there is a certain degree of attention between them, which enables them to mutually focus on each other in the process of information aggregation. In addition, we observe that distant nodes in the syntactic dependency graph can also be highly concerned when information is aggregated if there is an important semantic connection between them. For example, there is a high attention between “allele” and “panitumumab”, even though they are not adjacent to each other and are far away in fact. In summary, under the guidance of syntactic information, SR-GAT network can help each node to capture the information of other nodes that are far away but have semantic connections with each other, effectively learn the syntactic structure of long sequences, and has significant advantages in processing cross-sentence text.

5.3.4. Error analysis

We performed error analysis to the experimental results on the test set to detect the causes of model prediction errors. One case where we found prediction error instances was that, there were multiple entities in cross-sentence and there were interfering relations between some of the entities, and at the same time, the relation was not explicitly mentioned between the entities involved in the relation to be extracted. For

example:

“cell UNK the mutant egfr are in general more sensitive to tkis than cell express the wild-type kinase. in particular, the l858r mutant is 10- to 100-fold more sensitive to *erlotinib* and gefitinib than the wild-type kinase, and significantly more sensitive than the *g719s* mutant.”.

Where bold italic words indicate the entities involved in the relation to be extracted. In this example, the fact that “l858r mutant is sensitive to erlotinib and gefitinib drugs” is mentioned, so the model will predict the relation as “sensitivity”, while the target entities involved in the relation to be extracted, “g719s” and “erlotinib”, are not explicitly mention the relation, so actually the correct relation should be “None”. Therefore, the predicted relation “sensitivity” is actually not correct.

In addition, we report a statistical analysis of model prediction errors in Fig. 4. According to statistics, the test set contains 1173 relation instances. The number of predicted errors in the “Drug-Mutation” binary five-classification relation extraction task is 83, and the number of predicted errors in the “Drug-Gene-Mutation” ternary five-classification relation extraction task is 115. Further, we note that most errors occur in the prediction of the “None” relation. In particular, about 30% of error instances are prediction errors for the “resistance or non-response” and “None” relations. After analyzing these error examples, we found that expressions of these two types of relationships are consistently unclear and require pharmacology background knowledge. This means that even manual annotation may result in errors.

For the first type of error, we consider that, the prediction results are influenced by the interfering relations between other entities in the sentences due to two reasons that, the SR-GAT network over-understands the meanings expressed in the sentences, and the representation of entities’ information is not perfect because of the simply averaging operation. In future work, improvements in fusing sentence structure features and entities features can be considered, and these improvements can further enhance the performance of our model in the cross-sentence relation extraction task.

5.3.5. Feature representation for other tasks

The SR-GAT network is essentially a feature engineering network, and the above experiments show that it performs well in the cross-sentence relation extraction task. To further demonstrate that the feature representations learned by SR-GAT network are effective and can be transferred well to other tasks effectively, we present its performance in comparison with other advanced models for the text classification task in Table 4. In extending our method to the text classification task, we simply adjust the downstream task network, i.e., we use the

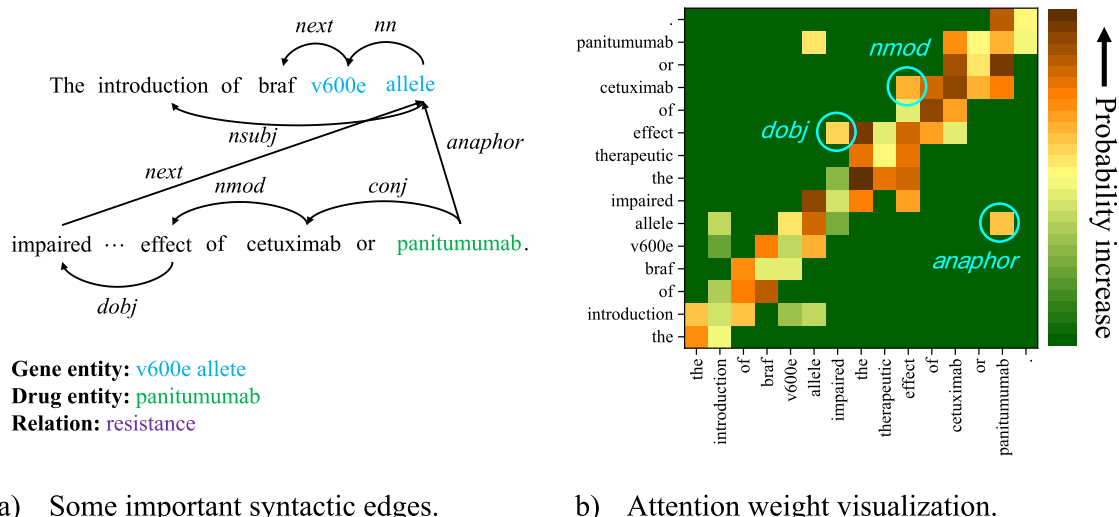
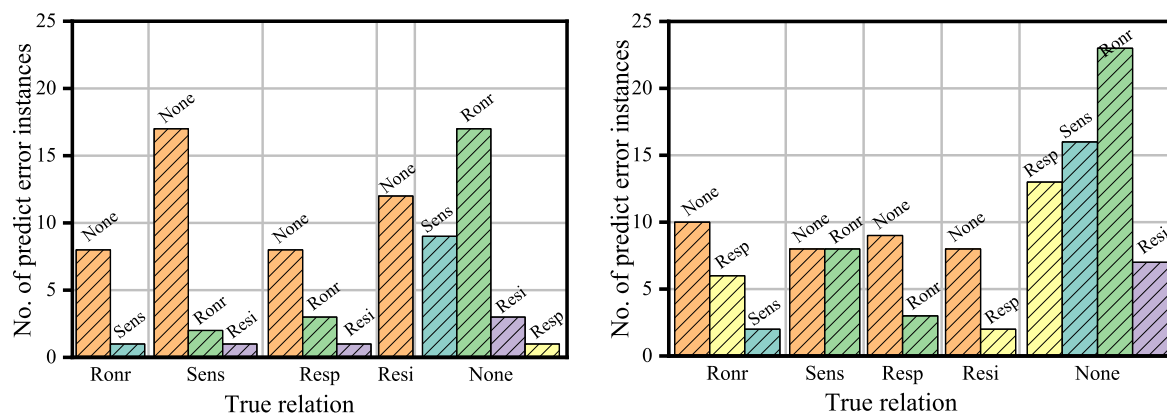


Fig. 3. An example for “The introduction of braf v600e allele impaired the therapeutic effect of cetuximab or panitumumab.”



a) Distribution of prediction error examples for the "Drug-Mutation" binary five-classification relation extraction task.

b) Distribution of prediction error examples for the "Drug-Gene-Mutation" ternary five-classification relation extraction task.

Fig. 4. Prediction error examples analysis figure. The Y-axis in the Fig. 4 indicate the number of instances where the true relation is predicted to be the wrong relation. Among them, "Ronr" denotes "resistance or non-response", "Sens" denotes "sensitivity", "Resp" denotes "response", "Resi" denotes "resistance" and "None" denotes "None".

Table 4

Performance comparison of different methods on the CoLA dataset. Bold number in the table indicates the best score.

Methods	Accuracy (%)
DistilBERT (Sanh et al. [30], 2019)	49.1
SqueezeBERT (Iandola et al. [31], 2020)	46.5
SpanBERT (Joshi et al. [32], 2020)	64.3
XLNet (Yang et al. [33], 2020)	69.0
STRUCTBERT (Wang et al. [34], 2020)	69.2
ALBERT (Lan et al. [35], 2020)	69.1
FLOATER-large (Liu et al. [36], 2020)	69.0
24hBERT (Izsak et al. [37], 2021)	57.1
Charformer-Tall (Tay et al. [38], 2022)	51.8
Vector-wise (Dettmers et al. [39], 2022)	68.6
FNet-Base (Lee-Thorp et al. [40], 2022)	69.0
FNet-Large (Lee-Thorp et al. [40], 2022)	78.0
SR-GAT (Ours)	69.5

feature representations learned by the SR-GAT network as the input to the classification network directly.

As shown in Table 4, our method lags behind the FNet-Large by 8.5% accuracy, but outperforms other methods. The accuracy of most methods concentrates between 60% and 70%, and one of the reasons for this result is the unbalance of positive and negative samples in the CoLA dataset. We find that most of the pre-trained language model (LM) based methods were able to achieve more than 65% performance, confirming its extraordinary performance in migration learning. However, the excellent performance of LM depends heavily on the parameters of the model and the scale of the pre-trained corpus, which makes it challenging to train and generalize. For example, the DistilBERT trained a smaller linguistic representation model by using less data, and although they introduced a knowledge distillation strategy in the pre-training phase, the final performance of the model was still very limited, with an accuracy of only 49.1%. The 24hBERT used fewer model parameters in order to reduce pre-training costs, although they used a combined strategy of software optimization, design selection, and hyperparameter tuning to achieve an accuracy of only 57.1% in this task. In addition, we note that although the FNet-Large achieves the highest accuracy, its base

version, the FNet-Base, has an accuracy of only 69.0%. In comparison, the number of parameters for the FNet-Large is about four times larger than that of the FNet-Base (83 M for the FNet-Base compared to 238 M for the FNet-Large), which confirms that more parameters and training data are required for the LM based methods to achieve better performance.

In summary, our method has a simple structure compared to the LM based methods and achieves a competitive performance, which shows that the feature learned by the SR-GAT network is effective and it can be robustly generalized to other tasks without additional fine-tuning.

6. Conclusion and future work

In this paper, we propose a novel graph attention network guided by syntactic dependency (SR-GAT) for extracting relations between multiple entities from cross-sentence text. Because cross-sentence text often has too long sequences, it is difficult for sequence-oriented modeling methods to learn effective feature representation. The attention mechanism of graph structure-oriented modeling can effectively solve this problem, where each node can pay attention to the information of other nodes in its neighborhood without the limitation of sequence length. Different from the traditional GAT network, the SR-GAT network aggregates the feature representation of neighborhood nodes with the guidance of the SR-GPR network, which encodes the syntactic dependency between nodes and can effectively learn the dependency structure information of long sequences. The experimental results show that our method achieves state-of-the-art performance in the biomedical cross-sentence relation extraction task. In addition, our method achieves reasonable performance in the text classification task, further showing that the feature representation learned by the SR-GAT network is effective and can be applied to other domains successfully. Overall, our method has a simple structure and can learn global dependency information in long sequences well, which has significant advantages in dealing with long sequence problems and can help researchers quickly and accurately mine biomedical knowledge from the vast amount of literature.

The limitation of our method is the use of an independent external parser to obtain the syntactic dependency graph. It is obvious that it

generates a large number of task-irrelevant or redundant relations, especially when applied to a specific domain, and invalid relations can even cause wrong focus between words to affect the performance of the model. In future work, we will seek a pruning strategy to complete the refinement of the syntactic dependency graph to obtain the best sentence structure relevant to the task, further improving the performance of our method and making it easier to be generalized to any domains.

CRedit authorship contribution statement

Xueyang Zhou: Conceptualization, Methodology, Software, Software, Writing – original draft, Investigation. **Qiming Fu:** Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing, Supervision. **Jianping Chen:** Conceptualization, Resources, Writing – review & editing, Supervision. **Lanhui Liu:** Validation, Writing – review & editing. **Yunzhe Wang:** Validation, Writing – review & editing. **You Lu:** Validation, Writing – review & editing. **Hongjie Wu:** Validation, Writing – review & editing.

Appendix A. The operation mechanism of different methods

As can be seen from Fig. A1 (a), the calculation of recursive network is time-dependent, and the calculation of the current time needs to use the output of the previous time. As a result, its computation cannot be parallelized and it has low computational efficiency when dealing with long sequence problems. As can be seen from Fig. A1 (b), the self-attention network allows more parallelization, and the computation of each time step can be carried out simultaneously, which greatly improves the computational efficiency. However, when the sequence is too long, the computational complexity of the self-attention mechanism will be significantly increased, and the dense connection of all nodes will make it difficult for the model to learn long-term dependent information. Further, as shown in Fig. A1 (c), SR-GAT (ours) uses syntactic dependency graph to guide the aggregation of information in the attention mechanism, so that each node only needs to focus on nodes within its neighborhood, which significantly reduces the complexity of model learning.

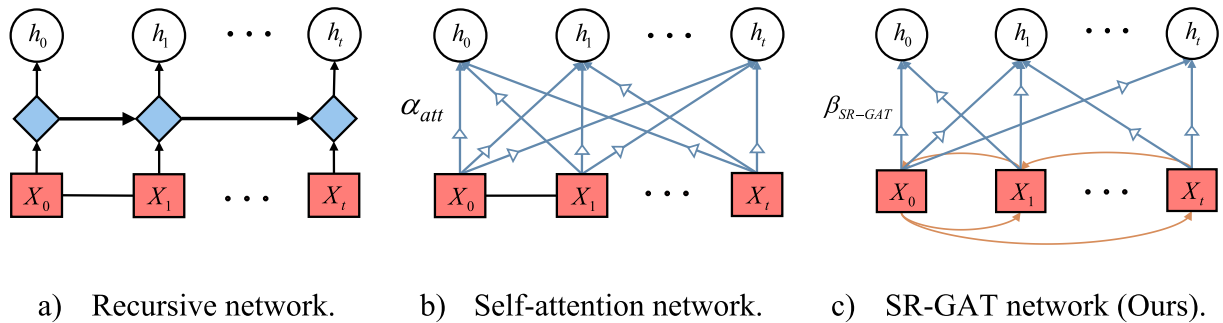


Fig. A1. Modeling principles of different methods.

Appendix B. The overall architecture of our method

See Fig. B1

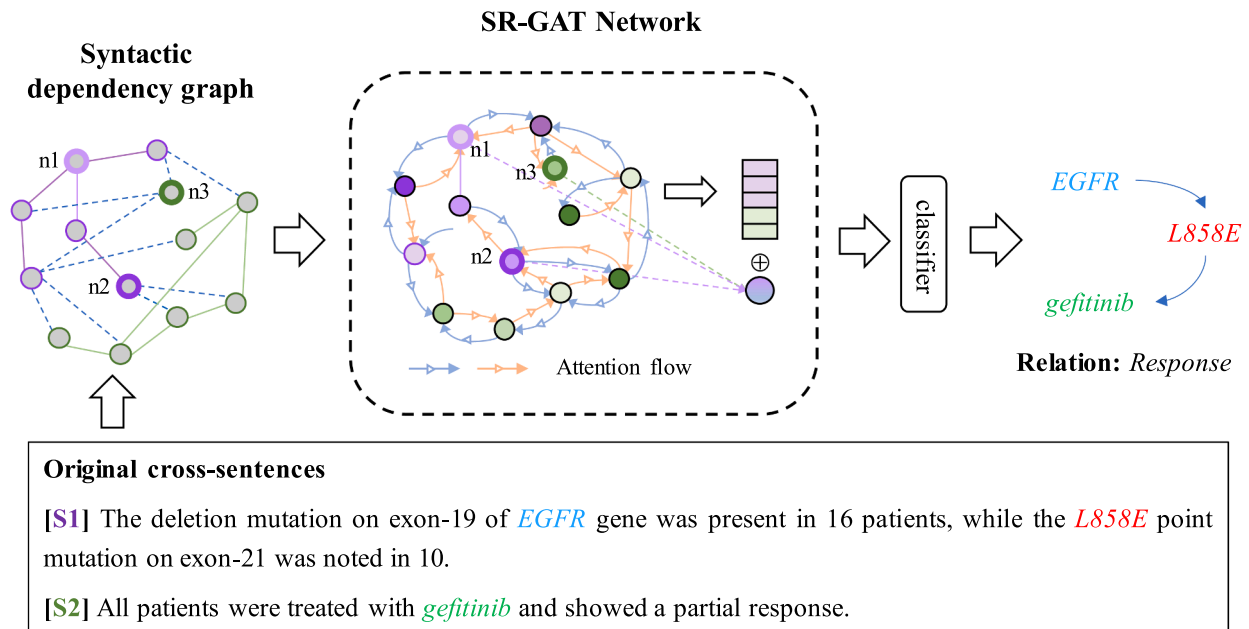


Fig. B1. The overview of our model. In the syntactic dependency graph, we use purple and green to represent sentences 1 and 2, respectively, and n_i to represent the entities needed to extract relation. In the SR-GAT network. Meanwhile, we use $\oplus$ to represent the direct concatenation of feature vector and entities vector.

Appendix C. The example of syntactic dependency graph

As shown in Fig. C1, It can be seen that, the relation between “EGFR-gene”, “l858e-mutation” and “point-mutation” is “NN”, which indicates that they are common nouns. The relation between “treated-gefitinib” is “PREP_WITH”, which indicates the use of the modifier of the preposition “with”, i. e., the treatment with “gefitinib”. In addition, for a more detailed description of the relation, please refer to Marneffe et al. [41].

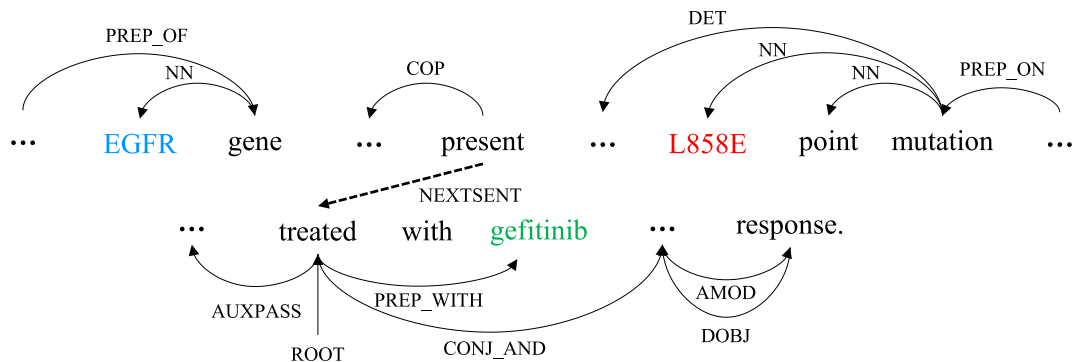


Fig. C1. Syntactic dependency graph for the example: “The deletion mutation on exon-19 of *EGFR* gene was present in 16 patients, while the *L858E* point mutation on exon-21 was noted in 10. All patients were treated with *gefitinib* and showed a partial response.”. For simplicity, we show only part words in the two sentences and their dependencies, and omit the adjacency between adjacent words. In Fig. C1, blue indicates the gene entity, green indicates the drug entity, and red indicates the mutation entity.

Appendix D. The task network

See Fig. D1

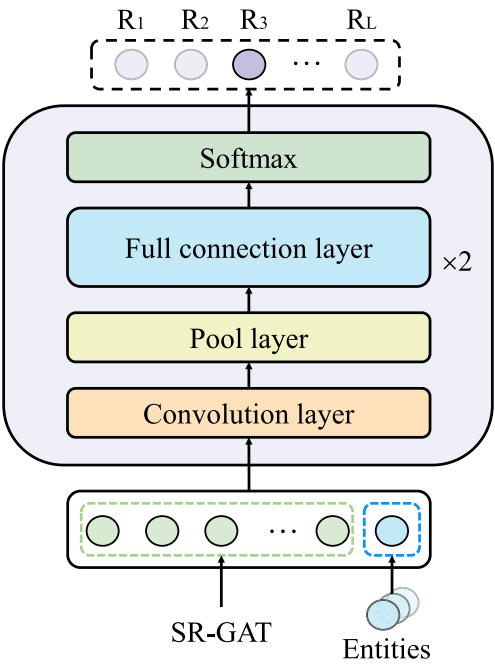


Fig. D1. Task Network.

Appendix E. Experimental parameter settings

See Table E1

Table E1	
Experimental parameter settings.	
Parameters	Value
Word embedding dimension	100
Edge embedding dimension	50
Maximum sentence length	128
Hidden state dimension in fully connected layers	512
Number of parameters space in SR-GPR network	3
Dropout probability	0.6
Learning rate	0.0001

Appendix F. The performance distribution of our model

We further report performance ranges (min, median, max) of our method in five-fold cross-validation, and the results are shown in Fig. F1 The scatters in the figure represent the performance distribution of other existing methods, and the boxplot is the performance distribution of our method in five-fold cross-validation. As can be seen from Fig. F1, the performance of our model is relatively stable, and the results of the five-fold cross-validation are distributed in a small range. In addition, we can see that even the worst performance of our model is significantly competitive.

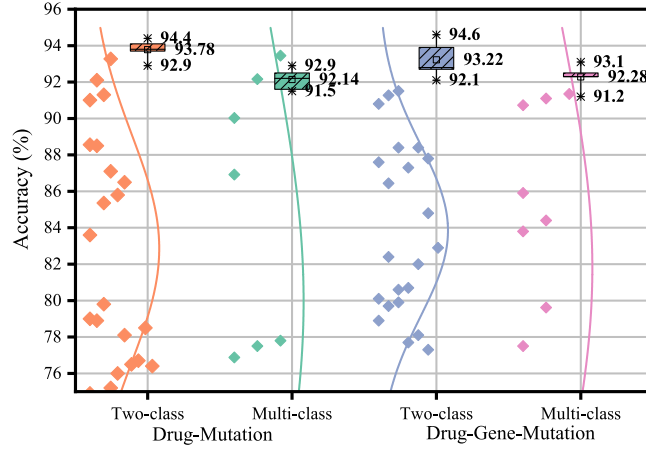


Fig. F1. Distribution of five-fold cross-validation results.

Appendix G. Formulaic modeling

In order to describe the implementation details of our method more clearly, we formulate the whole execution process of the model in this section.

We formulate the cross-sentence relation extraction task as follows: Given an input document: $S = \{s_i\}_{i=1}^n$ and a labeled entity set: $E = \{e_i\}_{i=1}^m$, mapping them to a specific relation in the predefined relation set $\mathcal{R} = \{R_i\}_{i=1}^{|\mathcal{R}|}$ to detect the relation mentioned in the text by the entities.

In this paper, we first analyze the syntactic dependency of the input document $S = \{s_i\}_{i=1}^n$ by using the Stanford Parser, and obtain a syntactic dependency graph $\Omega = (\mathbb{N}, \Psi)$, where $\mathbb{N} \in \{w_i\}_{i=1}^k$ and $\Psi \in \{\varphi_i\}_{i=1}^t$ are the set of words in sentences and the set of syntactic edges between them, respectively. We then feed it into SR-GAT to compute the deep feature representation. In SR-GAT, the syntactic dependency graph Ω is preprocessed to obtain an edge set $L_\Omega \in \mathbb{R}^{K \times K}$ and a mask matrix $M_\Omega \in \mathbb{R}^{K \times K}$, where K is the maximum of node. In particular, we construct virtual edges ('pad') to extend them to dimension K for nodes that have no syntactic dependencies. We use the a 2D convolution layer and a 2D max-pooling layer to extract the feature of the edge set L_Ω , as Eq. (1).

$$Output_{full} = W * \text{Maxpool}(\text{Conv}(\text{Ebd}(L_\Omega))) + b \quad (1)$$

Where, "*" is the matrix multiplication and $\text{Ebd}(L_\Omega)$ represents the word embedding. W and b are learnable parameters. Furthermore, we mask the correlation coefficients calculated by the virtual edges, as Eq. (2).

$$Output_{mask} = M_\Omega \cdot Output_{full} \quad (2)$$

Where, dot "." represents the element-wise multiplication of the matrix. Finally, we use the softmax function to transform the correlation coefficient matrix into a correlation probability matrix, as Eq. (3).

$$P_\Omega = \{p_1, p_2, \dots, p_K\} = \text{Softmax}(Output_{mask}^i)_{i \in \{1, 2, \dots, K\}} \quad (3)$$

Where p_i is a K -dimensional vector that represents the correlation weights between the i -th node and all nodes in the sentence.

In addition, for $\mathbb{N} \in \{w_i\}_{i=1}^k$, we transform it to the vector representation by word embedding. For any node w_i , it will be transformed as follows.

$$v_i = \text{Ebd}(w_i) \quad (4)$$

Further, we multiply it with the probability matrix to obtain the deep feature of node, as Eq. (5).

$$h_i = p_i * v_i \quad (5)$$

Where p_i is the i -th row in the correlation probability matrix P_Ω , representing the correlation scores between node w_i and all nodes in the sentences. For the whole input syntactic dependency graph, its features can be expressed in Eq. (6).

$$H_\Omega = P_\Omega * V_\Omega \quad (6)$$

Where H_Ω is the learned feature representation of Ω , and V_Ω is the word embedding of $\mathbb{N}$. Further, we compute features using multiple networks with independent parameters, and aggregate them as the output of SR-GAT using an average method, as Eq. (7).

$$H_{output} = 1/N \sum_{i=1}^N (H_\Omega^i) \quad (7)$$

Where N is the total number of parameter spaces and H_Ω^i is the feature representation learned by the i -th set of parameters.

In the task network, we first extract the shallow feature of the entities through the average method, as Eq. (8).

$$F_E = 1/m \sum_{i=1}^m v_i \quad (8)$$

We then concatenate it with the output of the SR-GAT network as the input of the task network, as Eq. (9).

$$F = H_{\Omega} \oplus F_E \quad (9)$$

Where, $\oplus$ is the direct concatenation of vectors. It will be mapped to the predefined set $\mathfrak{R}$ of relations after the following transformation.

$$P\mathfrak{R} = \text{softmax}(W_2 * (\tanh(W_1 * \text{Maxpool}(\text{Conv}(F_{all}))) + b_1)) + b_2) \quad (10)$$

Where $P\mathfrak{R}$ is the output of the task network and its dimension is the number of relations in the pre-defined set of relations $\mathfrak{R}$, which is the predicted probability for each relation.

Appendix H. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jbi.2023.104445>.

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